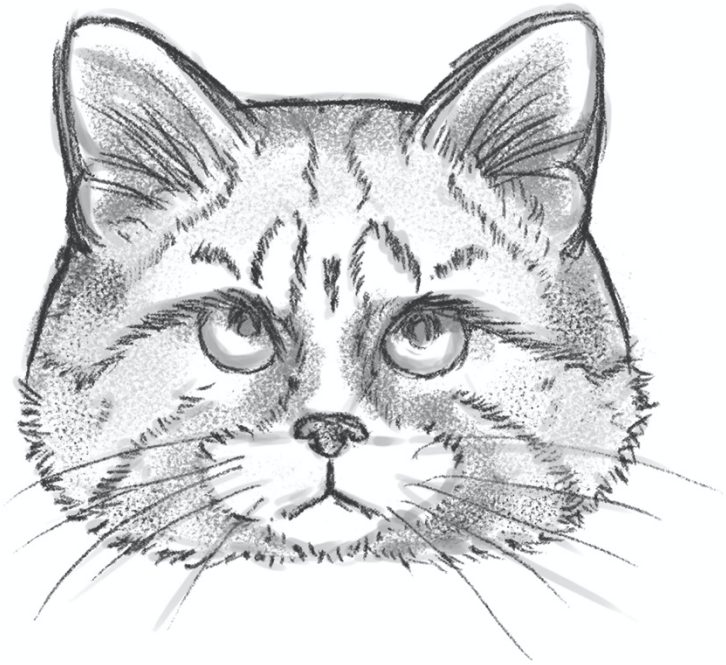
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FÜR BIODIVERSITÄT UND NACHHALTIGE LANDNUTZUNG

– CENTRE OF BIODIVERSITY AND SUSTAINABLE LAND USE –

The effect of anthropogenic disturbances on local wildlife: a comparative study of German wildcat populations



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Abstract

Natural populations experience an increasing amount of anthropogenic disturbance as human development continues. Central Europe underwent a severe drought in 2018, which has been related to anthropogenic climate change. As climatic anomalies are expected to become more common, their effects on natural populations are important to investigate. Elusive species such as the European wildcat (*Felis silvestris silvestris*) are commonly monitored by the use of camera trapping. In the present study, 20 camera trapping sites in Kaufunger Wald (KW) and 25 camera trapping sites in Melsunger Bergland (MB) were used to estimate abundance and population density of the European wildcat in a disturbed (KW) and less-disturbed area (MB). The potential impact on wildcats caused by anthropogenic disturbance was further explored by comparing the relative abundance indices from MB and KW. To investigate how future camera trapping of the European wildcat can be improved, GLMs were fitted to investigate the effect of various habitat characteristics and human activity on capture rate. As in the previous study by Wening et al. (2019), none of the habitat characteristics were found to explain variation in wildcat capture rate, and neither did the parameters related to human disturbance. However, 44/45 camera trapping sites successfully detected the European wildcat, which support reliability of the study design. The European wildcat was photocaptured at similar rates in both MB and KW, which was further supported by the similarity of the density estimates from both areas obtained from conventional capture-recapture and spatially explicit capture-recapture. Thus, none of the methods support a difference in density between disturbed KW and the less-disturbed MB. Consequently, the present study conclude that the European wildcat is fairly tolerant to anthropogenic disturbance.

Introduction

Anthropogenic forces such as climate change and related disturbances are important drivers of change in natural systems (Parmesan and Yohe 2003, Rosenzweig et al. 2008). Recent biological trends e.g. range boundary shifts and changes in species abundance can be attributed to anthropogenic climate change (Parmesan and Yohe 2003, Rosenzweig et al. 2008). Concurrently, the effect of human-mediated habitat alterations has led to local extinctions and an overall decline in the global biodiversity (Barnosky 2009). Wildlife monitoring is therefore vital to insure the initiation of appropriate wildlife conservation measures (Butchart et al. 2010).

The spring and summer of 2018 was characterized by concurrent seasonal anomalies in temperature and precipitation, resulting in an extreme drought unprecedented for Central Europe (Toreti et al. 2019). The event caused high mortality in many tree species as a result of extreme water stress and secondary drought effects (Schuldt et al. 2020). Especially, Norway spruce (*Picea abies*), a common species in cultivated forests, was highly affected. The drought made the species more susceptible to infestations by the Eurasian spruce bark beetle (*Ips typographus*) during and after the event (Schuldt et al. 2020). The seasonal anomalies have been linked to anthropogenic climate change and similar drought conditions are expected to become common with the current climate change projections (Toreti et al. 2019, Vogel et al. 2019). Therefore, understanding their potential impact on natural populations is a current concern.

The consequences of the 2018 drought are also evident in the area considered in the present study, where the elevated tree mortality in the Norway spruce followed by the removal of the dead trees has impacted forest structure and composition. As the data collection in Kaufunger Wald (KW) was conducted in 2019, the data reflects an area disturbed by anthropogenic climate change in comparison to the data collected in 2017 in the relatively less disturbed Melsunger Bergland (MB). Furthermore, wind energy development is currently taking place in KW, and the forest contained a total of 10 wind turbines at the time of data collection. While direct mortality, e.g. from collisions, is regarded as the primary impact to wildlife stemming from wind turbines, indirect responses such as altered behaviour or displacements can negative impact natural populations (Northrup and Wittemyer 2013). Very little research has been conducted on non-volant species in this regard, but localized anthropogenic noise or other disturbances related to construction and maintenance of the wind turbines are likely to impact local populations (Northrup and Wittemyer 2013, Jerem and Mathews 2020). KW, therefore, represents an area where anthropogenic disturbance could have had an impact on the local animal community.

The European wildcat (*Felis silvestris silvestris*, Schreber 1777) is a solitary mesocarnivore belonging to the Felidae family (Anile et al. 2018). The species was once widely distributed across Europe, but extensive persecution in the 18th and 19th century and continuing habitat loss has led to local extinctions and strongly reduced abundance (Piechocki 1990). Five genetically distinct populations are currently recognized; two of which inhabit Central and Western Germany, respectively (Mattucci et al. 2016). However, a study by Steyer et al. (2016) suggests that current range expansion has reconnected the German wildcat populations. The European wildcat is associated with broadleaf and mixed forest systems, where they utilize dense vegetation structures to rest (Wittmer 2001, Klar et al. 2008). However, the higher prey density at the forest edge and in nearby

fields makes them optimal feeding grounds for the wildcat (Kilshaw et al. 2015, Beugin et al. 2016). Often regarded as a rodent specialist, the European wildcat feeds opportunistically on the small animals available within their home-range, but tend to specialize more in rabbits, when they are present (Malo et al. 2004).

Despite being categorized as 'Least Concern' by the IUCN, the European wildcat currently experience an overall population decline (www.iucnredlist.org, 2020). The species is known to avoid human settlements and related anthropogenic features e.g. roads and heavily used trails (Piñeiro et al. 2012, Oliveira et al. 2018), although males might be more tolerant to anthropogenic disturbances than females (Klar et al. 2008). Regionally, the species experiences persecution in the shape of non-selective predator control (Virgós and Travaini 2005, Gil-Sánchez et al. 2020). As the species is known to be sensitive to anthropogenic features (e.g. roads and settlements), other structures such as e.g. wind turbines could potentially; directly or indirect through the effect on their prey, negatively impact the fitness of the species (Klar et al. 2008, Northrup and Wittemyer 2013). As fragmentation and low density is seen as some of the major challenges faced by the European wildcat, on-going anthropogenic expansion and energy development could be a major threat to the species (Northrup and Wittemyer 2013, Mattucci et al. 2016, Gil-Sánchez et al. 2020).

Camera trapping is considered an efficient non-invasive method for detecting the presence of rare species or estimate the abundance of wildlife (Karanth and Nichols 1998, Rovero and Zimmermann 2016). This method is considered cost and labour efficient compared to other detection methods (e.g. line transects, track surveys or telemetry), while simultaneously causing a minimum of environmental disturbance (Silveira et al. 2003, Balme et al. 2009, Rovero and Zimmermann 2016). The cameras operate day and night for extended periods of time, which makes them suited to study elusive species (Rovero and Zimmermann 2016). In the past decade, camera trapping has become a widely used method to estimate abundance and density of natural populations (Rovero and Zimmermann 2016). Camera trapping is particularly applicable to species, where individuals can be identified by natural marks (e.g. fur patterns like spots and stripes), as the individual capture histories can be analysed in a capture-recapture framework (Karanth 1995, Carbone et al. 2001, Rovero and Zimmermann 2016). Consequently, the method has been widely applied to monitoring of felids (e.g. tiger *Panthera tigris*; Karanth and Nichols 1998, puma *Puma concolor*; Kelly et al. 2008, leopard *Pantera pardus*; Allen et al. 2020, margay *Leopardus wiedii*; Horn et al. 2020), including the European wildcat (Anile et al. 2010, Kilshaw et al. 2015).

The number of photocaptures per sampling effort, known as the photographic capture rate (PCR), is sometimes used as an index of relative abundance, especially in species that cannot be individually identified by natural markings (Rovero and Marshall 2009, Palmer et al. 2018). PCR is strongly related to density; a pattern, which has also been found for the European wildcat (Rovero and Marshall 2009, Gil-Sánchez et al. 2020). However, capture-recapture analyses requires an sufficient amount of data (e.g. photocaptures) to provide reliable abundance and density estimates (Amstrup et al. 2005). Therefore, the trapping site should be optimized to increase detection probability of the target animal, which can be improve through knowledge of habitat preferences and natural movement patterns (Rovero and Zimmermann 2016, Wening et al. 2019). Radio-tracking has revealed, that the European wildcat utilize the forest boundary, meadows and open areas in the forest, while anthropogenic structures such as road and settlements are typically avoided (Klar et al. 2008).

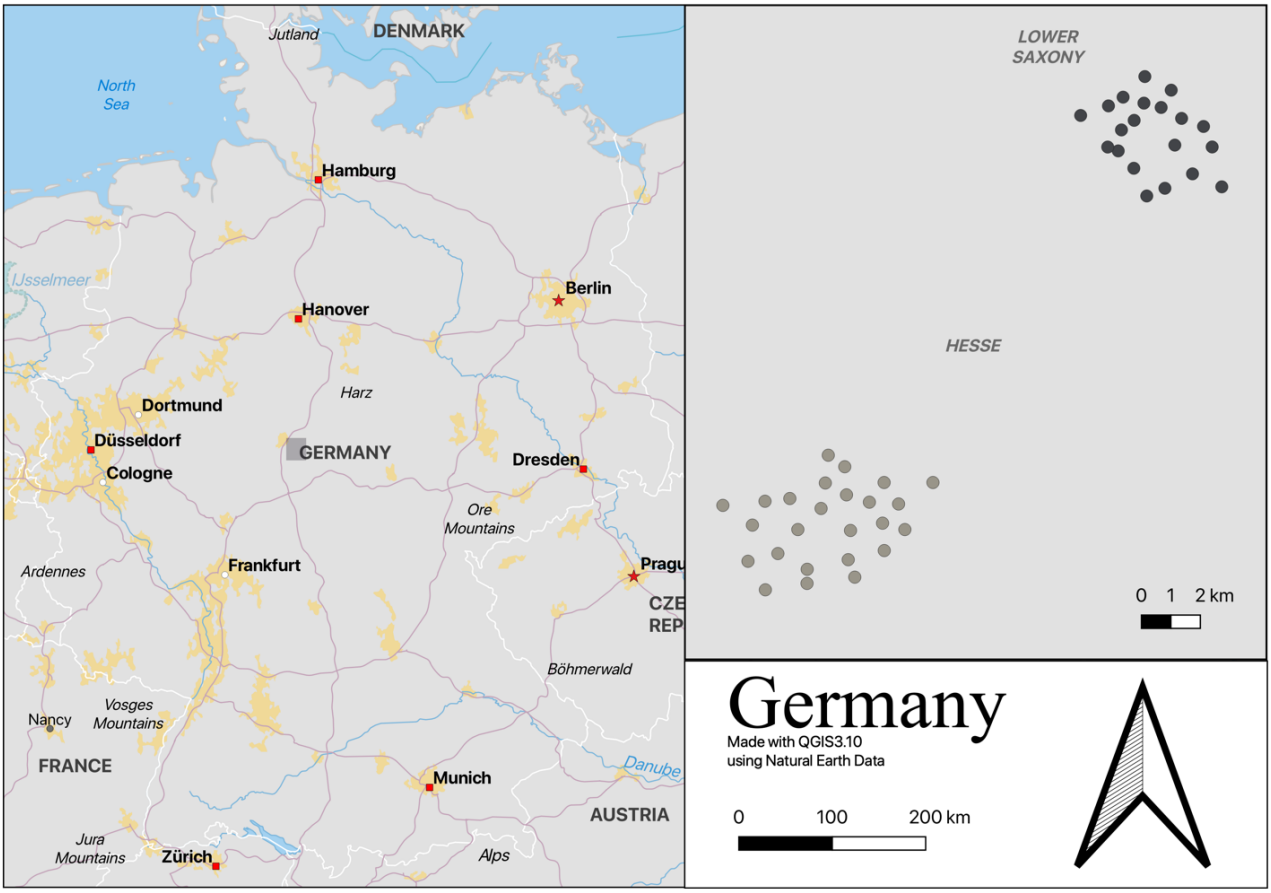


Figure 1: Schematic image of Germany on the left-hand side. Red lines mark major roads. The grey square indicates the location of the study area, which is highlighted on the right-hand side. The light grey dots indicate the trapping sites in MB, and the dark grey dots indicate the trapping sites in KW.

Wening et al. (2019) analysed the data from MB to investigate, which site characteristics were associated with high PCR. The findings were inconclusive, which could be due to the relatively low sample size (N=25). As the present study evaluates 45 trapping sites, the opportunity arises to re-visit the research question in order to optimise future camera trapping success.

The present study aims to: (i) Estimate European wildcat (*Felis silvestris silvestris*) abundance and population density using conventional and spatially explicit capture-recapture analyses; and to (ib) Compare these estimates between a strongly disturbed area and a less disturbed area. In relation to this, the study aims to: (ii) Compare the relative abundance derived from the photographic capture rate between a strongly disturbed area and a less disturbed area. Finally, to re-address the findings by Wening et al. (2019), the study aims to: (iii) Assess the impact of various habitat characteristics on photographic capture rate of the European wildcat (*Felis silvestris silvestris*) and to: (iiib) Assess the impact of human activity on photographic capture rate.

Methods

Study areas

The studies were carried out in the federal state of Hesse, Germany, in MB and KW (Figure 1). The two study areas are located approximately 15 kilometres away from each other. They are characterized by forest cover, mainly a mix of coniferous and broadleaf forest of varying age, and they support diverse animal communities. Both MB and KW are used for recreational activities e.g. hiking, horse-back riding and mountain biking, while simultaneously being used for timber production. Furthermore, the study area in KW contains a total number of 10 wind turbines.

Set in a low mountain range, the two study areas experience a mild and humid climate. The mean daily temperatures range between -1°C and 24°C and the mean monthly precipitation range between 1.5 mm/day and 2.6 mm/day corresponding to Central European weather conditions (https://www.worlddata.info/europe/germany/climate-hessen.php). The data collection was carried out in 2017 and 2019 for MB and KW, respectively. Central Europe experienced an extreme drought event in the year between the two data collections, which i.a. ruined many spruce trees (*Picea abies*) in the study areas (Toreti et al. 2019). Consequently, this provides an opportunity to investigate potential effects of this type of disturbance on the European wildcat.

Habitat characteristics

The habitat surrounding each camera trapping site was described using three different characteristics: forest type, age of forest stand and trail type. Forest type was broadly separated into three different categories (broad-leaved, coniferous and mixed) based on the dominant vegetation type within a 100-meter radius of the location of the cameras.

The age of the forest stand was evaluated within a 100-meter radius from the camera trapping site and divided into three age groups: early successional stage, stem exclusion and mature forest. The first category included low vegetation characterized by densely positioned young trees and bushes. The second spanned from trees in early development to initiation of storey stratification, while the third category consisted of mature trees with well-developed stratification (modified from Wening et al. 2019).

Trail type was separated into two types: forest trail and road. The two were roughly distinguished by the accessibility for and frequency of use by motorized vehicles on the latter. Aerial images were processed using ArcMap 10.5.1 (Esri, Inc., Redlands, CA, USA) to determine the distance from the camera trapping sites to the nearest constant water source (creek, pond or ditch) and the nearest forest boundary. A forest boundary was defined as any 'outer' forest edge (typically to villages, fields or roads) or as the boundary to a clearing within the forest with a minimum area of 20m * 20m.

Camera trapping

The camera trapping survey in MB ran from June 26 to October 8 in 2017 (i.e. 104 days), and the survey in KW ran from March 18 to July 19 in 2019 (i.e. 123 days). A 1 * 1-km grid was overlaid the two study areas, omitting the city Kehrenbach and surrounding fields in MB. The size of the grid was selected to avoid that a full wildcat home range could fit into a gap between trapping sites, which ensure that all individuals have an encounter probability > 0 (Karanth and Nichols 1998, Rovero and Zimmermann 2016). A trapping site was placed in each grid cell at forest roads or trails to optimize capture probability (Carbone et al. 2001, Wening et al. 2019). 25 camera sites were placed in MB with an average spacing of 846.54 meters, whereas 20 sites were placed in KW with an average spacing of 671.62 meters.

A pair of movement and heat-activated digital cameras were placed perpendicular to the trail or road to obtain pictures of both flanks for optimal identification (Karanth 1995, Rovero and Zimmermann 2016). The cameras used xenon white flash to obtain colour images both day and night and were set with the shortest possible time between consecutive images. The cameras were placed

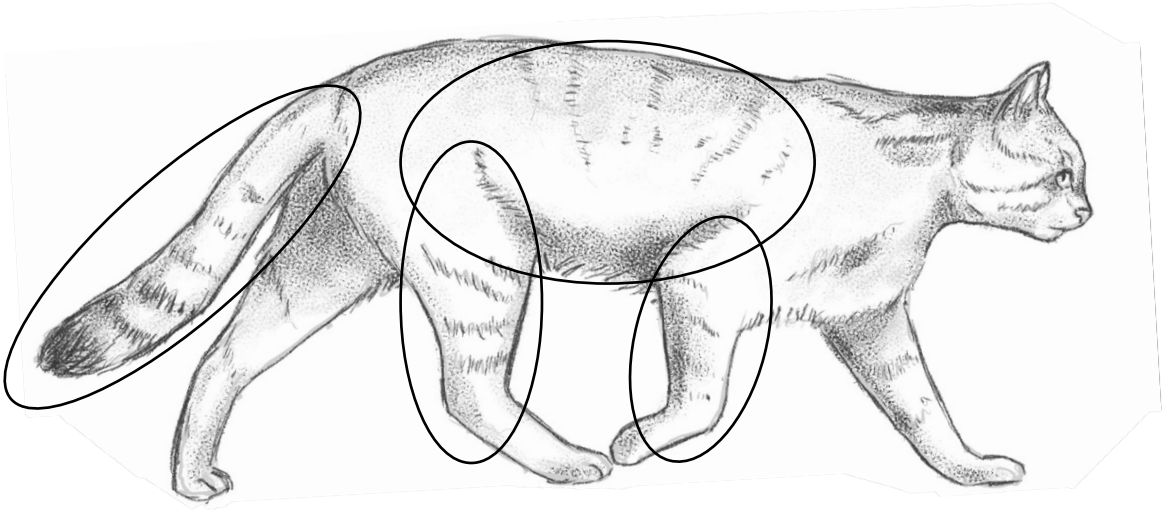


Figure 2: Illustration of a European wildcat. The circles mark areas on the wildcat that are more variable between individuals and therefore contain patterns that might serve to identify individual wildcats. © Katrine Brandstrup

on poles or trees approximately 30 centimetres above the trail or road at a distance from one another to avoid flash interference (Karanth and Nichols 1998). They were secured using metal boxes locked with a padlock or cable lock. The camera traps used were Cuddeback (Cuddeback Digital, Green Bay, USA) of the models Ambush, Attack and C1 in MB and the models C1 and G in KW, and they were checked every 2-4 weeks to cut down intervening vegetation and replace batteries and SD cards (Rovero and Zimmermann 2016).

Individual identification

Individual identification was carried out by a total of three observers – two observers per study area to reduce misidentification errors (Rovero and Zimmermann 2016). All detections of cats with typical wildtype characteristics have been included. Some sources deem distinguishing European wildcat from domestic cat x wildcat hybrids possible from pelage characters alone (Ragni and Possenti 1996, Anile et al. 2010), other sources disagree (Krüger et al. 2009, Kilshaw et al. 2015). Therefore, the accidental inclusion of hybrids cannot be rejected.

Individual identification was done by assigning an image of a new individual as reference image for that individual. Subsequent images were then compared to a reference image or groups of images of already known individuals. European wildcats are individually identifiable from pelage characters (Anile et al. 2010). The present study primarily used fur patterns on flank, tail and outer side of both legs to distinguish individual wildcats from one another (Figure 2). One particular individual was, in addition, recognizable by its white toes on both hindlegs.

The camera traps obtained a total of 219 (113 of the left flank, 106 of the right flank) and 156 (76 of the left flank, 80 of the right flank) images of European wildcats in KW and MB, respectively. Prior to analyses, 23 (20.35%) left flank images and 16 (15.09%) right flank images from KW and 10 (13.16%) left flank images and 20 (25.00%) right flank images from MB were discarded. The excluded pictures were of lower photo quality or displayed the cat at an angle or only in part,

obscuring essential features for identification. The remaining images were used for individual identification.

The images were both observers agreed that the cat was identifiable were used for photographic capture-recapture. The individual identifications yielded three datasets from each study area for further analyses. One from each observer and one concordant dataset, where only the images that both observers agreed upon were included.

Conventional capture-recapture

The different datasets obtained from individual identification were used to create a capture history matrix (see Appendix 1). In a capture history a photocapture of a particular individual is marked by a '1' in the occasion of detection, while '0' is used to mark that the individual was not encountered in that particular occasion (Amstrup et al. 2005, Rovero and Zimmermann 2016). To keep the probability of violating the closed-population assumption as low as possible, the sampling periods were shortened to run from March 25 to June 16 (84 days) for KW and July 2 to October 7 (98 days) for MB. The study periods were divided into one-week and two-weeks sampling occasions, resulting in 7 and 14 occasions for MB and 6 and 12 occasions for KW. Hence, each dataset was analysed twice; once divided into one-week occasions and once divided into two-week occasions.

Capture histories were created for each dataset and utilized to estimate abundance using the package 'Rcapture' in Rstudio (version 1.2.5019, R version 3.6.1, Rstudio Team 2020). The function 'closedp' was used to fit various loglinear models (M0, Mt, Mh Poisson2, Mth Poisson2, Mb and Mbh) that test the data for heterogeneity (h), time (t) and behavioural effects (b). The models were evaluated using the Akaike's information criteria (AIC), which estimates the relative amount of information lost due to uncertainty. Hence, the lowest value indicates the best supported model. All models with $\Delta AIC < 2$ were considered, as this generally suggests substantial support for the model (Rovero and Zimmermann 2016).

Density estimates were obtained by dividing the abundance estimates from the best supported models with the size of the investigated area (km^2). Three methods were used to calculate buffer width to estimate the size of the effectively sampled areas: 1) home range radius (HRR); estimated to 1.11km by Brozovic 2016 (unpublished BA thesis) from the average home range of 30 individual wildcats estimated using radio telemetry, 2) mean maximum distance moved by individuals seen on >1 occasions (MMDM) and 3) half mean maximum distance moved (HMMDM). MMDM and HMMDM were calculated independently for KW and MB. The buffer width was added as a boundary strip to the minimum convex polygon (MCP) marked by the outer traps (Karanth and Nichols 1998). The MCPs were calculated with the percentage parameter set to 100. In addition, density was also estimated using the MCP without a buffer.

Spatially explicit capture-recapture

In addition to the density estimates found using conventional capture-recapture (CCR), densities were estimated with spatially explicit capture-recapture (SECR) using the package 'secr' in Rstudio (version 1.2.5019, R version 3.6.1, Rstudio Team 2020). SECR is more complex than CCR and circumvents the element of calculating the effectively sampled area by incorporating the movement

and distribution of individuals in space in relation to the trapping sites (Rovero and Zimmermann 2016).

The SECR analyses required an encounter history and a table with coordinates of all camera trapping sites (Rovero and Zimmermann 2016). The encounter histories were created using both the left and the right flank concordant datasets from MB and KW separately. Each row in the table contains the sessions, wildcatID, occasion and trapping site of one successful photocapture (see Appendix 2). The encounter histories were prepared in order to use the detector type 'proximity', meaning that only one entry per individual per occasion per site is permitted (Rovero and Zimmermann 2016). SECR modelling required a mask, which was added to the camera trapping sites as a buffer. To create the mask, the 'root pooled spatial variance' (RPSV) was calculated, which is the two-dimensional dispersion of the detection locations pooled over the individuals (Rstudio Team 2020). The mask was then fitted as 6*RPSV and with a spacing of 500.

Three different models were fitted for each encounter history. A M0-model (null-model), which assumed constant values for animal density, baseline encounter probability and the parameter sigma. The second model investigated individual heterogeneity by treating the baseline encounter probability as a 2-part finite mixture, while keeping animal density and sigma constant. The last model treated both the baseline encounter probability and the parameter sigma as 2-part finite mixtures. As with the CCR models the SECR models were evaluated using AIC.

Analyses of relative abundance: General linear models

The PCR for both MB and KW was calculated as the total number of photocaptures divided by sampling effort. PCR is expected to be higher in an area with a larger number of wildcats (Carbone et al. 2001, Gil-Sánchez et al. 2020). Thus, to further investigate a potential difference in the number of wildcats present in KW and MB, a general linear model (GLM) with negative binomial error structure and log link function was fitted. The model included the number of wildcat detections per trapping site as a response variable, the area and path type as explanatory variables and the sampling effort, measured as the no. of days the cameras at each trapping site were functioning, as an offset term.

An independent detection was defined as any photocapture of a European wildcat separated by another photocapture by a minimum of 5 minutes. When both cameras obtained pictures of the same event, this was regarded as one detection. Independent detections of European wildcat obtained from all camera trapping sites were used to investigate the relationship between trapping success and a range of site and habitat specific variables. In addition to the target species, cameras also photocaptured humans (and pets) and cars (and other motorized vehicles). The number of photos of each category (humans and cars) per site was noted, and the site-specific capture rates were calculated by dividing the number of photos in each category by the sampling effort at each site.

To re-evaluate the findings from Wening et al. 2019, the additional 20 capture sites KW were added to the analyses. Preliminary analyses suggested, that there was no measurable difference between the two study areas, for which reason the following models did not include study area as a random term. Two negative binomial GLMs with log link function were fitted. The model was fitted with the various habitat characteristics as explanatory variables with the number of wildcat detections as response variable and sampling effort as an offset term. The second model was fitted with the

human activity related variables (humans and cars) as explanatory variables with the number of wildcat detections as response variable and sampling effort as an offset term.

The analyses were performed using the package 'MASS' in RStudio (version 1.2.5019, R version 3.6.1, Rstudio Team 2020).

Results

Individual identification

Mismatches between the observers' identifications excluded 14 (left flank) images and 11 (right flank) images from KW from further analyses. The interobserver agreement was 84,44% and 87,78% for left and right flank, respectively, which resulted in a concordant dataset with 14 individuals identified from both flanks. The following truncation of the data reduced these numbers to 12 and 11 individuals for the left and right flank, respectively. For the images collected in MB, mismatches between the observers' identifications led to the exclusion of 2 images (left flank) and 16 (right flank) images, which resulted in interobserver agreements of 96,67% for left flank images and 73,33% for right flank images. The concordant dataset included 19 individuals identified from the left flank and 18 individuals identified from the right flank.

Conventional capture-recapture

The abundance estimates obtained from analyses of the data divided into one-week occasions generally provided similar results to those divided into two-week occasions, therefore only the one-week occasions will be considered going forward. Likewise, estimates obtained from the datasets from independent observers were comparable to each other and to the estimates obtained from the concordant datasets. Some estimates obtained from datasets of single observers were slightly higher than estimates obtained from datasets using concordant identification ratings. As concordant estimates are more reliable, because a single observer is more prone to make errors, only estimates from concordant identifications will be reported in the next sections (Kelly et al. 2008). Estimates obtained from single observer data are provided in Appendix 3.

AIC supported some of the models including behavioural response, but this pattern was not consistently found in all analyses, making a "true" behavioural response unlikely. In particular, the behavioural response was typically only supported in the analyses of one flank, but not in the analyses of the other flank. Moreover, behavioural response is a particular variant of heterogeneity and can be difficult to discriminate from individual heterogeneity (Amstrup et al. 2005, Royle et al. 2014). Therefore, models including behavioural response have been excluded from further analyses.

The concordant one-week models from KW with $\Delta AIC < 2$ were M0 ($\Delta AIC = 0$) and Mh poisson2 ($\Delta AIC = 1.38$) from the right flank dataset and M0 ($\Delta AIC = 0$) and Mh poisson2 ($\Delta AIC = 1.21$) from the left flank dataset. The weighted average abundance estimate was 11.1 independent wildcats for the right flank models and 12.24 independent wildcats the left flank models. The concordant one-week models from MB with $\Delta AIC < 2$ were M0 ($\Delta AIC = 0$), Mt ($\Delta AIC = 0.28$), Mth poisson2 ($\Delta AIC = 1.81$) and Mh poisson2 ($\Delta AIC = 1.87$) for the right flank identifications, providing a weighted average abundance estimate of 20.84 independent wildcats. The models with $\Delta AIC < 2$ from the left flank identifications were Mt ($\Delta AIC = 0$) and Mth poisson2 ($\Delta AIC = 1.46$), resulting in a weighted average of 20.1 independent wildcats (Table 1).

Table 1: The AIC supported conventional capture-recapture models (ΔAIC< 2) estimated abundance from concordant one-week datasets in Kaufunger Wald (KW) and Melsungen Bergland (MB). SA = Study Area; ΔAIC = Relative AIC to the best model; RF = Right flank; LF = Left flank; Akaike w_i = Akiake weight; SE = Standard Error; WAA = Weighted Average Abundance.

SA	Model	AIC	ΔAIC	Akaike w_i	Abundance (±SE)	WAA
KW_{RF}	M0	138.15	0	0.66	11.1 (± 0.4)	
	Mh poisson2	139.53	1.38	0.33	11.1 (± 0.4)	11.1
KW_{LF}	M0	141.33	0	0.64	12.2 (± 0.5)	
	Mh poisson2	142.54	1.21	0.35	12.3 (± 0.6)	12.24
MB_{RF}	M0	151.44	0	0.38	21.0 (± 2.3)	
	Mt	151.72	0.28	0.33	20.6 (± 2.1)	
	Mth poisson2	153.25	1.81	0.15	20.7 (± 2.4)	
	Mh poisson2	153.31	1.87	0.15	21.1 (± 2.2)	20.84
MB_{LF}	Mt	171.57	0	0.67	20.1 (± 1.3)	
	Mth poisson2	174.03	1.46	0.32	20.1 (± 1.3)	20.1

The weighted average abundance estimates were utilized to calculated density estimates using the various buffers added to the area specific MCP and the unbuffered MCP (Table 2). In KW density using the MCP was estimated to 1.07 wildcats/km^2 and 0.97 wildcats/km^2 from left and right flank datasets, respectively. The estimates obtained with HRR as an added buffer were 0.41 wildcats/km^2 and 0.37 wildcats/km^2 for left and right flank datasets, respectively. The estimates obtained using MMDM and HMMDM were 0.33 wildcats/km^2 and 0.54 wildcats/km^2 for the left flank datasets and 0.30 wildcats/km^2 and 0.49 wildcats/km^2 for the right flank datasets. In MB the estimated density ranged from 0.45 wildcats/km^2 (HRR), through 0.47 wildcats/km^2 (MMDM) and 0.65 wildcats/km^2 (HMMDM) to 0.98 wildcats/km^2 (MCP) from the left flank datasets. The density estimates from the right flank datasets ranged from 0.47 wildcats/km^2 (HRR), through 0.49 wildcats/km^2 (MMDM) and 0.68 wildcats/km^2 (HMMDM) and 1.02 wildcats/km^2 (MCP). The densities from KW did not vary significantly from the densities obtained from MB ($P > 0.05$, Pearson's Chi-squared test, df = 7).

Spatially explicit capture-recapture

The AIC evaluation of the SECR analyses supported the model, which treated both the baseline

Table 2: Density estimates (wildcats/km^2) obtained by dividing the weighted average abundance estimates from the AIC supported conventional capture-recapture models (ΔAIC< 2) with different estimates of the effectively sampled area. Data from concordant one-week datasets in Kaufunger Wald (KW) and Melsungen Bergland (MB). RF = Right flank; LF = Left flank; MCP = Maximum convex polygon; HRR = Home range radius; MMDM = Mean Maximum Distance Moved; HMMDM = Half Mean Maximum Distance Moved. HRR, MMDM and HMMDM were all used as buffers added to MCP.

Study Area	MCP	HRR	MMDM	HMMDM
KW_{RF}	0.97	0.37	0.30	0.49
KW_{LF}	1.07	0.41	0.33	0.54
MB_{RF}	1.02	0.47	0.49	0.68
MB_{LF}	0.98	0.45	0.47	0.65

Table 3: The AIC supported spatially explicit capture-recapture models from Kaufunger Wald (KW) and Melsungen Bergland (MB). Data from concordant encounter histories individually identified from left and right flank detections. RF = Right flank; LF = Left flank; SE = Standard Error; CI = Confidence Interval.

Study Area	Model	Density	SE	Lower CI	Upper CI
KW_{RF}	D~1 g0~h2 sigma~h2 pmix~h2	0.40	0.25	0.12	1.12
KW_{LF}	D~1 g0~h2 sigma~h2 pmix~h2	0.43	0.17	0.20	0.89
MB_{RF}	D~1 g0~h2 sigma~h2 pmix~h2	0.88	0.25	0.50	1.53
MB_{LF}	D~1 g0~h2 sigma~h2 pmix~h2	0.73	0.28	0.35	1.52

encounter probability and the parameter sigma as 2-part finite mixtures in all cases (Table 3). In KW the analyses obtained density estimates (±SD) of 0.40 (±0.25) wildcats/km^2 from the right flank identifications and of 0.43 (±0.17) wildcats/km^2 from the left flank identifications. These estimates were comparable to the estimates obtained from CCR were a buffer was added to the MCP. The density in MB was estimated to 0.88 (±0.25) wildcats/km^2 and 0.73 (±0.28) wildcats/km^2 for the right and left flank identifications, respectively. These estimates were higher than the density estimates obtained through the CCR analyses using MCP + the different buffers to estimate the effectively sampled area.

General linear models

The camera traps operated a total of 2332 days and 2552 days in KW and MB, respectively. They accumulated a total of 168 (KW) and 164 (MB) detections of European wildcats, corresponding to a PCR (±SD) of 7.25 (±5.28) and 6.43 (±4.47) for KW and MB, respectively. Average no. of events per sites (±SD) were 8.4 (±6.00) and 6.56 (±4.69) for KW and MB, respectively. The target species was photocaptured at all camera trapping sites in MB, and in 19 out of 20 sites (95%) in KW.

The GLM investigating a potential difference in capture rate between the two study areas and path type did not significantly explain any variability. The capture rate of European wildcat overall was slightly lower in MB, but the difference between study areas was not significant (GLM estimate (MB) = -2.69, SE = 0.23, z value = -0.576, P value = 0.57). Additionally, capture rate was similar on roads and forest trails in both study areas (GLM estimate (Roads) = 0.17, SE = 0.23, z value = 0.76, P value = 0.45; Figure 3).

Table 4: Results of the GLM relating the capture rate of European wildcat to habitat characteristics. Path type (Forest), broadleaved forest and age group 1 are represented by the intercept. SE = Standard Error; CI = Confidence Interval.

	Estimate	SE	Z-value	P value	Lower CI	Upper CI
Intercept	-2.42	0.44	-5.46	<0.001	-3.27	-1.55
Path type (Road)	0.12	0.23	0.49	0.62	-0.34	0.57
Coniferous forest	-0.47	0.64	-0.74	0.46	-1.70	0.85
Mixed forest	-0.12	0.29	-0.42	0.67	-0.69	0.44
Age group 2	-0.21	0.28	-0.75	0.45	-0.74	0.34
Age group 3	-0.20	0.32	-0.62	0.54	-0.81	0.43
Distance to water	0.00007	0.0006	0.12	0.91	-0.001	0.001
Distance to forest boundary	-0.0006	0.0006	-0.82	0.41	-0.002	0.0008

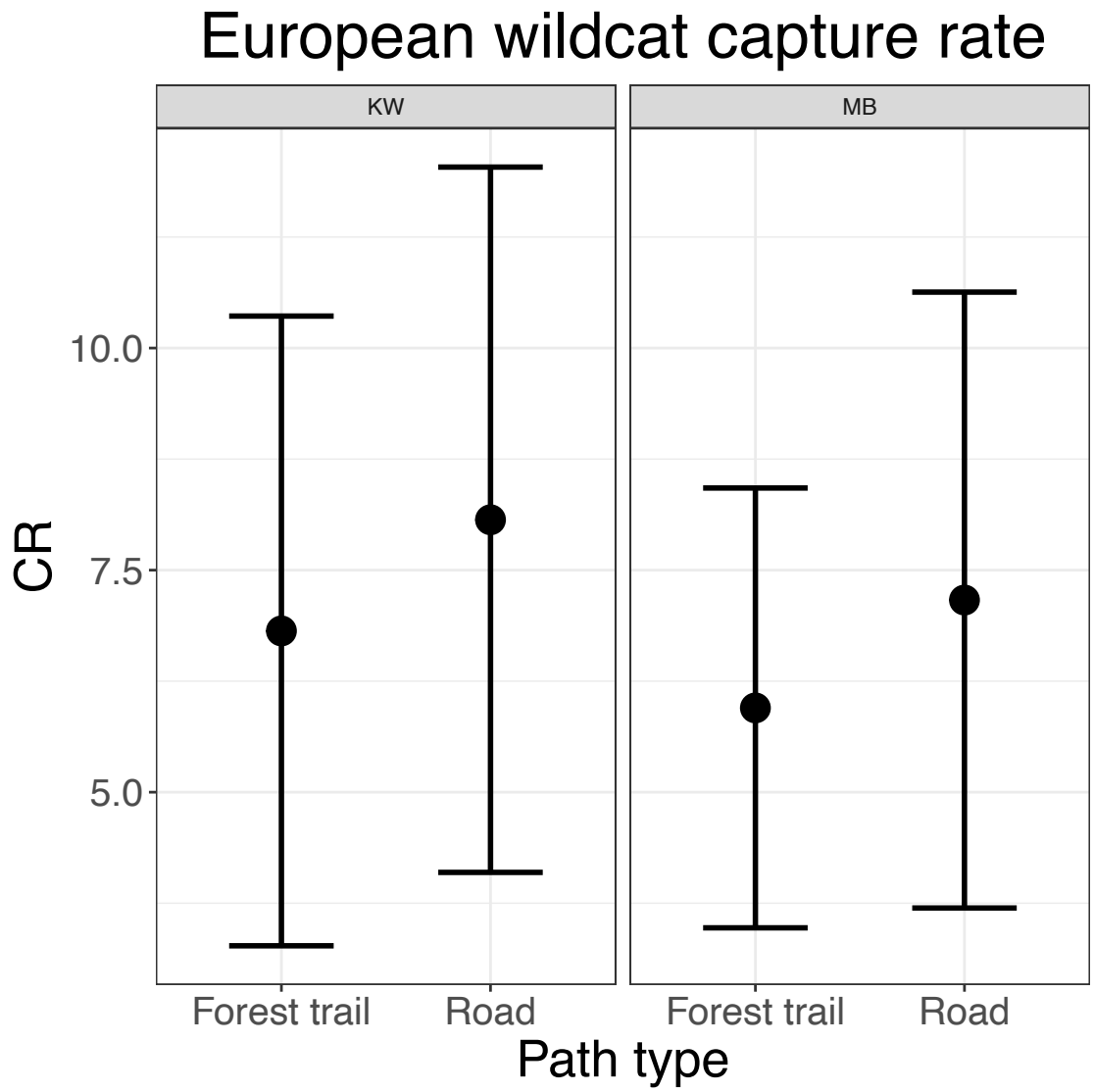


Figure 3: Plot including whiskers of the capture rate in KW and MB on forest trails and road, respectively.

Two GLMs were fitted to re-investigate the findings from Wening et al. (2019). The GLM relating the capture rate of European wildcat to habitat characteristics did not find that any of the investigated habitat characteristics has a significant effect on capture rate (Table 4). The GLM investigating the effect of human disturbance on capture rate did likewise not find any significant effects (Table 5). These results support the findings from Wening et al. (2019) by replicating the same results with 20 additional capture sites.

Discussion

In the present study, camera trapping was performed in two sampling areas in Hessen, Germany to investigate the impact of anthropogenic disturbance on the European wildcat. Abundances and population densities were estimated using conventional and spatially explicit capture-recapture

analyses. The resulting estimates did not suggest a significant difference between the disturbed KW compared to the less-disturbed MB. This was further supported by the PCR, which was similar between the two study areas. Additionally, the present study aimed to optimise future camera trapping efforts by assessing, whether various habitat characteristics and human activity had an impact on PCR. None of the parameters had a significant effect, which support the original conclusion reached by Wening et al. (2019).

The quality of the images obtained of the European wildcat were generally high, and the majority of the detections could be used for individual identification. The interobserver agreement for left and right flank identifications in KW were comparable, while both the highest and lowest interobserver agreement related to the data collected in MB. The difference in interobserver agreement from MB is expected to be caused by a time constraint, which gave one of the observers less opportunity to thoroughly carry out the right flank identifications. The level of disagreement between observers in respect to the KW data was attributable mainly to a collection of images that one observer identified as one wildcat, while the other observer identified two different individuals. This disagreement caused almost half of the mismatches between observers on both right flank detections and left flank detections. Misidentification errors can result in an overestimation of abundance, when known individuals are marked as new individuals, but the use of two observers reduce such errors and increase the reliability of the data (Kelly et al. 2008, Foster and Harmsen 2012, Rovero and Zimmermann 2016).

The density estimates obtained from CCR varied from 0.30 wildcats/km^2 to 0.54 wildcats/km^2 in KW and from 0.45 wildcats/km^2 to 0.68 wildcats/km^2 in MB with the three different buffers used. As evident in the present study, the density estimate vary greatly with the method used for calculating the effectively sampled area (Balme et al. 2009, Pesenti and Zimmermann 2013). Therefore, to estimate density the approximation of the effectively sampled area should be as close to the actual area used by the target species as possible (Karanth and Nichols 1998, Balme et al. 2009). The HMMDM buffer was originally proposed by Karanth and Nichols (1998), and while it has been critiqued for overestimating the population density estimate compared to MMDM, the general census suggests, that HMMDM provides the more reliable density estimates of the two (Parmenter et al. 2003, Balme et al. 2009, Pesenti and Zimmermann 2013, Royle et al. 2014). However, telemetry is likely to provide the most reliably approximation of the effectively sampled area (Parmenter et al. 2003, Balme et al. 2009, Rovero and Zimmermann 2016). As telemetry was not included in the present study, the HRR buffer strip was obtained from the average home-range of 30 different radio-tracked individuals; executed by Brozovic (2016). MMDM and HMMDM are constrained by the spatial sampling design and are strongly impacted by low levels of recaptures, thus utilizing the HRR buffer strip circumvents these limitations (Parmenter et al. 2003).

The SECR method circumvents the issue of estimating the effectively sampled area entirely by modelling the movement and distribution of individuals relative to the trapping sites (Rovero and Zimmermann 2016). Utilizing this analysis provided density estimates (±SD) of 0.40 (±0.25) wildcats/km^2 and 0.43 (±0.17) wildcats/km^2 for KW. The SECR estimates were intermediate of the highest and lowest density estimates obtained using CCR, suggesting that the KW density estimates are reliable. Compared to similar studies (e.g. 0.32±0.1 wildcats/km^2 in Sicily, Italy; Anile et al. 2014, 0.68±0.1 wildcats/km^2 in Scotland; Kilshaw et al. 2015, 0.069±0.002 wildcats/km^2 in Spain; Gil-

Sánchez et al. 2020), the density of the European wildcat in KW is moderate to high, while the density estimates obtained for MB (0.88 (±0.25) wildcats/km^2 for the right flank and 0.73 (±0.28) wildcats/km^2 for the left flank) were some of the highest observed. However, the standard errors of the SECR estimates are fairly high, which makes the estimates less reliable. SECR is sensitive to low sample size and tend to be more imprecise under field conditions, which results in slight overestimations (Keiter et al. 2017).

The density estimates obtained in the present study were overall high. The density estimates from CRR using the HHR buffer were in agreement with the estimates from SECR in KW. Contrastingly, the SECR estimates from MB were almost double the density estimates from CRR using the HHR buffer. As the SECR estimate has a large SE and would be the highest reported in literature for the European wildcat, its deemed unreliable. In consistency with the HHR estimates, the relative abundance is similar between study areas and the population density in the European wildcat has been previously been shown to significantly correlate with the PCR (Gil-Sánchez et al. 2020). The present study, consequently, suggests that the population density of the European wildcat has not been significantly affected by the higher anthropogenic disturbance in KW and the effect of the 2018 drought.

Heterogeneity in capture-recapture analyses can stem from both a behavioural response and individual heterogeneity, and the two causes are difficult to disentangle (Amstrup et al. 2005). In the present study, models including the behavioural response were excluded, as consistent support for the pattern was not found. A behavioural response entails a change in capture probability following the initial capture (Amstrup et al. 2005). Cameras are known to be both visibly and audibly detectable by animals, which could induce a negative response to detection. The noise omitted by camera traps is expected to be negligible in field conditions (Meek et al. 2014), whereas the intense white light produced by flash-type (xenon white flash) used in the present study is more likely to frighten target species (Glen et al. 2013, Rovero and Zimmermann 2016). Feral cats have been observed to respond to camera traps with apprehension (Glen et al. 2013). However, specific investigation of the influence of flash-type on feral cats did not confirm a particular behavioural response to cameras with white flash compared to infrared-flash cameras, nor did they find that the cameras reduced redetection (Taggart et al. 2020). This is likely applicable to the European wildcat, which is supported by previous camera trap surveys of the species, as they do not suggest trap-shyness (Anile et al. 2010, Can et al. 2011).

Part of this study aimed to further investigate whether the PCR of wildcats is affected by various habitat characteristics, which was originally examined by Wening et al. (2019) in MB. The findings by Wening et al. (2019) were supported, as none of the various habitat characteristics re-evaluated in this study significantly affected PCR of the European wildcat. The same conclusion has previously been reached for forest type (Klar 2007). While former studies investigating habitat use of the European wildcat have identified forest presence with close vegetation and proximity to meadows and watercourses as important land-features (Wittmer 2001, Klar et al. 2008). In contrast to the present study, they investigated the spatial organization of the European wildcat using radio-tracking. In its essence, camera traps provide coarse-grained information about species presence in a particular area, where telemetry provides fine-scale information about habitat use that might not be replicable with camera trapping (Klar et al. 2008, Gil-Sánchez et al. 2020).

The GLM models investigating the PCR in regard to human activity did not support any effect of human presence on wildcat detection rate. Males of the European wildcat has been observed to be more tolerant of habitat fragmentation and human presence (Oliveira et al. 2018). As the study in KW was conducted while the European wildcat typically give birth and nurse their young, the detection rate of females could be lower compared to males (Kilshaw et al. 2015). While this could explain some of the observed pattern, the population in MB supported the same relationship and was not conducted in the same months. Thus, higher detection of the less disturbed males, is unlikely to explain the result. However, while the European wildcat is known to avoid larger roads, the pattern is not consistent in smaller road systems, and the species commonly use trails and forest roads that are also frequented by humans (Can et al. 2011, Steyer et al. 2018, Wening et al. 2019, Beugin et al. 2020). The non-significant influence of human activity on the wildcat could be explained by temporal niche differentiation. Human activity is typically highest during the day, while multiple studies of the European wildcat supports slightly higher activity by night (Wittmer 2001, Germain et al. 2008). However, this pattern was not supported by a study of the European wildcat from Central Italy (Anile et al. 2018). Human activity has a strong influence on wildlife and has in multiple cases increased nocturnality. As human populations expand, wildlife might not have the opportunity to seek spatial refuge (Gaynor et al. 2018). Instead, increased nocturnality could facilitate human-wildlife co-existence, thus investigation of a temporal shift in response to humans in the European wildcat could provide information as to how the species could adapt to increased human presence.

The European wildcat is hypothesized to be particularly vulnerable to anthropogenic climate change (Stefen 2015), but the present study does not suggest that population density nor detectability of the wildcats were impacted by the extreme drought in 2018 and the related increase in anthropogenic disturbance. Previous studies have reported, that density of the European wildcat most strongly relates to the presence of abundant prey and dense vegetation structure to seek refuge in (Lozano et al. 2003, Malo et al. 2004, Klar et al. 2008). Additionally, the wildcat hunts in the forest boundary and utilize a more varied landscapes (Lozano et al. 2003, Klar et al. 2008). While the drought and the following removal of dead trees has likely changed the forest structure it might not directly affect the wildcat at this scale. (Schuldt et al. 2020).

Conclusion

The present study successfully estimated abundance and population density of the European wildcat using camera trapping in two different study areas. The estimates obtained from both CCR and SECR did not support a difference in density between the disturbed KW and the less-disturbed MB. The pattern was further supported by the similarity of the PCRs between the two study areas. The results suggest, that the European wildcat is less vulnerable to anthropogenic disturbance as previously assumed. The present study additionally re-visited the study by Wening et al. (2019) by investigating the impact of various habitat characteristics and human activity on the PCR. The analyses did not suggest, that any of the parameters investigated had a significant effect on capture rate of the European wildcat. However, all camera trapping sites apart from one in KW successfully detected the target species. Hence, the study design described in the present study can be reliably used for monitoring of the European wildcat.

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Table of abbreviations

Abbreviation	Definitions
AIC	Akaike's information criteria
CCR	Conventional capture-recapture
GLM	General Linear Models
HMMDM	Half Mean Maximum Distance Moved
HRR	Home range radius
KW	Kaufunger Wald
MB	Melsunger Bergland
MCP	Minimum Convex Polygon
MMDM	Mean Maximum Distance Moved
PCR	Photographic capture rate
RPSV	Root pooled spatial variance
SECR	Spatially explicit capture-recapture

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Appendix 1: Example of a capture history.
Kaufunger Wald. Concordant dataset. Right flank.

Cat/Occasion	1	2	3	4	5	6	7	8	9	10	11	12
1	0	0	0	0	0	0	0	0	0	0	0	0
2	1	0	0	0	1	0	0	1	0	0	0	0
3	1	0	1	1	1	1	1	0	1	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0
5	1	1	1	1	1	1	1	0	1	0	0	0
6	0	0	0	0	0	0	0	1	0	1	1	0
7	0	0	0	0	0	0	0	0	0	0	0	0
8	0	1	0	0	1	0	1	0	0	0	0	0
9	0	0	0	0	1	0	1	0	0	0	0	0
10	1	0	1	1	0	0	0	0	0	1	1	0
11	0	0	1	0	0	0	0	0	0	0	0	1
12	0	0	0	0	0	0	1	0	0	0	0	0
13	0	0	0	0	1	0	0	1	1	1	0	0
14	1	0	0	0	0	1	0	1	1	0	0	0

Appendix 2: Example of an encounter history.
Kaufunger Wald. Right flank.

Session	ID	Occasion	trapID
1	1	1	Hes12
1	1	5	Hes12
1	1	8	Hes82
1	2	1	Hes12
1	2	3	Hes12
1	2	4	Hes75
1	2	3	Hes82
1	2	4	Hes82
1	2	6	Hes82
1	2	7	Hes82
1	2	9	Hes82
1	2	3	Hes83
1	2	5	Hes85
1	2	7	Hes85
1	4	3	Hes79
1	4	6	Hes82
1	4	1	Hes85
1	4	2	Hes85
1	4	3	Hes85
1	4	4	Hes85
1	4	1	Hes86
1	4	4	Hes86
1	4	6	Hes86
1	4	7	Hes86
1	4	1	Hes87
1	4	9	Hes87
1	4	1	Hes88
1	4	5	Hes88
1	5	8	Hes83
1	5	10	Hes83
1	5	11	Hes83
1	5	10	Hes84
1	7	2	Hes85
1	7	5	Hes85
1	7	7	Hes85
1	7	9	Hes85
1	8	5	Hes10
1	8	7	Hes87
1	9	1	Hes72
1	9	3	Hes72
1	9	3	Hes75
1	9	4	Hes75
1	9	10	Hes75
1	9	11	Hes75
1	11	12	Hes75
1	12	7	Hes76
1	13	5	Hes78
1	13	8	Hes88
1	13	9	Hes88
1	13	10	Hes88
1	14	6	Hes10
1	14	1	Hes86
1	14	8	Hes86
1	14	8	Hes87
1	14	9	Hes87

Appendix 3

The AIC supported conventional capture-recapture models (ΔAIC< 2) estimated abundance from the additional tests on individual observers 1-week and 2-week datasets and the 2-week concordant dataset. ΔAIC = Relative AIC to the best model; O1: Observer 1; O2: Observer 2; C = Concordant; RF = Right flank; LF = Left flank; 1 = one-week occasions; 2 = two-week occasions; Akaike w_i = Akiake weight; SE = Standard Error; WAA = Weighted Average Abundance.

Study area: Kaufunger Wald						
Dataset	Model	AIC	ΔAIC	Akaike w_i	Abundance (±SE)	WAA
C_{RF2}	M0	67.03	0	0.48	11.1 (± 0.4)	
	Mt	68.35	1.32	0.25	11.1 (± 0.3)	
	Mh poisson2	69.02	1.99	0.18	11.1 (± 0.4)	11.1
C_{LF2}	M0	69.79	0	0.46	12.4 (± 0.7)	
	Mh poisson2	70.74	0.62	0.29	12.7 (± 1.0)	12.52
$O1_{RF2}$	Mt	65.30	0	0.52	11.1 (± 0.2)	
	Mth poisson2	66.13	0.83	0.34	11.1 (± 0.3)	11.1
$O1_{RF1}$	M0	143.79	0	0.67	11.1 (± 0.3)	
	Mh poisson2	145.22	1.43	0.33	11.1 (± 0.3)	11.1
$O1_{LF2}$	M0	70.42	0	0.47	12.3 (± 0.5)	
	Mh poisson2	70.94	0.52	0.37	12.6 (± 0.9)	12.43
$O1_{LF1}$	M0	146.70	0	0.78	12.2 (± 0.4)	
$O2_{RF2}$	Mh poisson2	79.46	0	0.47	16.3 (± 1.5)	
	M0	80.41	0.95	0.29	15.5 (± 0.8)	15.99
$O2_{RF1}$	M0	164.20	0	0.61	15.40 (± 0.7)	
	Mh poisson2	165.16	0.95	0.38	15.40 (± 0.7)	15.40
$O2_{LF2}$	Mh poisson2	79.46	0	0.47	16.3 (± 1.5)	
	M0	80.41	0.95	0.29	15.5 (± 0.8)	15.99
$O2_{LF1}$	M0	162.14	0	0.60	15.40 (± 0.7)	
	Mh poisson2	163.00	0.86	0.39	15.50 (± 0.8)	15.44
Study area: Melsungen Bergland						
Dataset	Model	AIC	ΔAIC	Akaike w_i	Abundance (±SE)	WAA
C_{RF2}	Mth poisson2	90.76	0	0.53	21.9 (± 3.0)	
	Mt	91.86	1.1	0.30	20.5 (± 2.1)	21.39
C_{LF2}	Mth poisson2	84.26	0	0.77	22.5 (± 2.8)	
$O1_{RF2}$	Mth poisson2	100.57	0	0.65	31.3 (± 4.0)	
	Mt	101.94	1.37	0.33	29.2 (± 2.9)	30.60
$O1_{RF1}$	Mt	178.01	0	0.66	29.2 (± 2.8)	
	Mht poisson2	179.53	1.52	0.31	29.3 (± 2.8)	29.23
$O1_{LF2}$	Mht poisson2	84.26	0	0.77	22.5 (± 2.8)	
$O1_{LF1}$	Mt	175.16	0	0.68	20.1 (± 1.2)	
	Mth poisson2	176.68	1.52	0.32	20.2 (± 1.3)	20.13
$O2_{RF2}$	Mth poisson2	93.52	0	0.80	28.1 (± 4.1)	
$O2_{RF1}$	Mt	182.44	0	0.92	24.8 (± 2.1)	
$O2_{LF2}$	Mth poisson2	86.51	0	0.83	25.8 (± 3.4)	
$O2_{LF1}$	Mt	177.08	0	0.67	22.6 (± 1.5)	
	Mth poisson2	178.49	1.41	0.33	22.6 (± 1.5)	22.6

Declaration of independent work

I hereby confirm that I have written the attached thesis on my own and that I did not use any resources than those specified above. This work has not been previously submitted, either in the same or in a similar form to any other examination committee and has not yet been published.

Randers, 9/11-2020	
Place, Date	Signature